

Part1

Variable: Highest year school completed mother & Number of children

H0: The variables are not linearly related to each other.

H1: The variables are linearly related to each other.

Correlations

		Highest year school completed, mother	Number of children
Highest year school completed, mother	Pearson Correlation	1	-.229**
	Sig. (2-tailed)		<.001
	N	2089	2088
Number of children	Pearson Correlation	-.229**	1
	Sig. (2-tailed)	<.001	
	N	2088	2344

** . Correlation is significant at the 0.01 level (2-tailed).

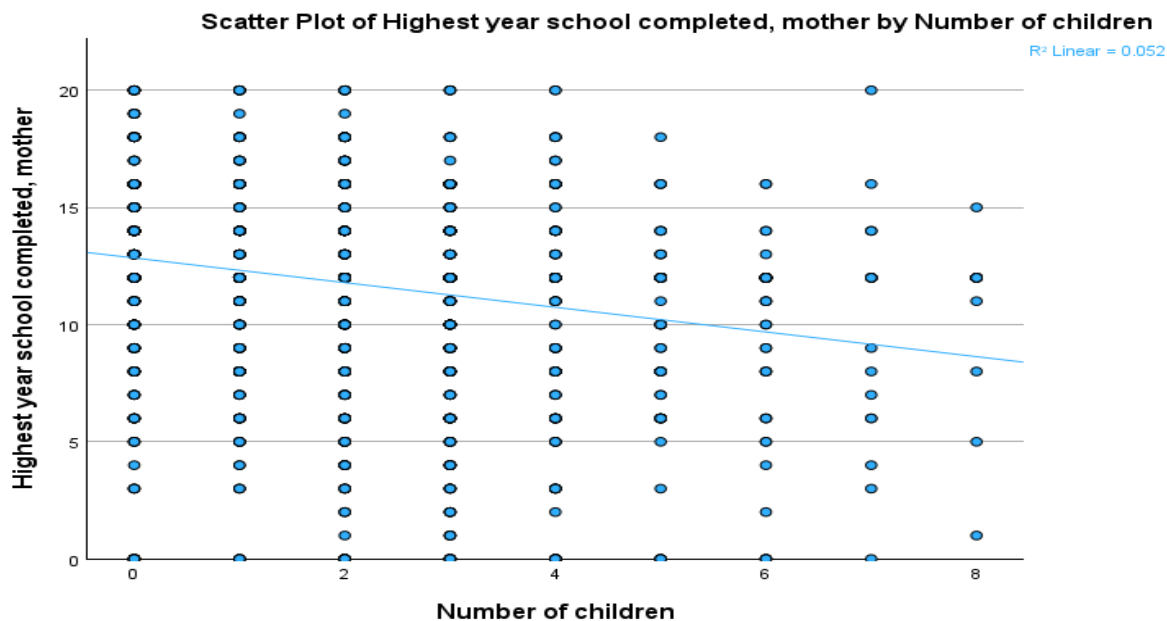
Result: A Pearson correlation test was conducted to examine the relationship between mother's highest year of school completed and number of children. The results showed a statistically significant negative correlation, $r = -0.229$, $p < .001$, indicating that mothers with higher levels of education tend to have fewer children. However, the strength of this relationship is weak. The sample size was large ($N = 2088$), which increases the reliability of the results.

Test Assumptions:

Both variables should be continuous variables. In this analysis, mother's highest year of school completed and number of children are treated as continuous variables, so this assumption is met.

The relationship between the two variables should be linear.

There are no influential outliers.



The scatterplot shows a negative downward slope and a weak linear relationship between mother's highest year of school completed and number of children.

Part 2

The purpose of this Pearson correlation test exercise studies the relationship between two continuous variables (Highest year school completed mother & Number of children), which is to determine if one variable increases in relation to another variable. The results showed a statistically significant negative correlation between the two variables, $r = -0.229$, $p < 0.001$, indicating that as a mother's level of education increases, the number of children tends to decrease. Although the relationship is relatively weak, it is statistically significant, suggesting a meaningful association between education level and family size in this sample. In addition, the sample size was large ($N = 2088$), which increases the reliability of the findings. Overall, the analysis suggests that higher maternal education is associated with having fewer children.